PHYSICS

Q1. A man stands in front of a mirror of special shape. He finds that his image has a very small head, a fat body, and legs of normal size. What can we say about the shapes of the three parts of the mirror?
a) Convex, Concave, Plane
b) The plane, Concave, Convex
c) Concave, Convex, Plane

ANSWER KEY SOON ON THIS CHANNEL

Q2. How ... if the diameter of the objective of ... telescope is doubled?
a) Two times b) Four times
c) Eight times d) Sixteen times

Q3. Which among the following is a portion of a transparent refracting medium bound by one spherical surface and the other plane surface?
a) Concave mirror b) Plane mirror
c) Lens d) Prism

Q8. During Einstein's Photoelectric Experiment, what changes are observed when the frequency of the incident radiation is increased?
a) The value of saturation current increases
b) No effect
c) The value of stopping potential increases
d) The value of stopping potential decreases

Q9. Which of the following did Bohr use to explain his theory?
... ation of mass

Q10. What is the mass of hydrogen in terms of amu?
a) 1.0020 amu b) 1.0180 amu
c) 1.0070 amu d) 1.0080 amu

Q11. The basic theorem/principle used to obtain mass-energy relation is __________
a) Heisenberg's Uncertainty Principle
b) Work-Energy Theorem
c) Momentum Conservation Theorem
d) Maxwell Theorem

Q4. X is thicker in the middle than at the edges, whereas, Y is thicker at the edges than in the middle. Identify 'X' and 'Y'.

a) X = concave lens; Y = convex lens
b) X = convex lens; Y = concave lens
c) X = plane lens; Y = convex lens
d) X = concave lens; Y = plane lens

Q5. The speed of yellow light in a certain liquid is 2.4×10^8 m/s. Find the refractive index of the liquid.

a) 1.25 b) 5.55
c) 6.25 d) 12.25

Q6. In Young's Double Slit Experiment, if instead of monochromatic light white light is used, what would be the observation?

a) The pattern will not be visible
b) The shape of the pattern will change from hyperbolic to circular
c) Colored fringes will be observed with a white bright fringe at the center

d) Maxwell Theorem

Q12. The manifestation of the band structure in solids is due to which of the following?

a) Heisenberg's uncertainty principle
b) Pauli's exclusion principle
c) Bohr's correspondence principle
d) Boltzmann's law

Q13. The band gap between the valence band and conduction band is the measure of ______

a) The conductivity of the material
b) The resistivity of the material
c) Charge density
d) Ease of ionization

Q14. If the positive terminal of the battery is connected to the anode of the diode, then it is known as:

a) Forward biased b) Reverse biased
c) Equilibrium d) Schottky barrier

Q15. When voltage is applied across cathode to anode ... said to be ______ biased.

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d) The bright and dark fringes will change position

Q7. If the frequency of the incident radiation is equal to the threshold frequency, what will be the value of the stopping potential?

a) 0
b) Infinite
c) 180 V
d) 1220 V

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Q15. When voltage is applied across cathode to anode then it is said to be ______ biased.

a) Reverse
b) Forward
c) Cyclic
d) Backward

Q16. A junction or a point where two or more network elements intersect is called

a) node
b) branch
c) loop
d) mesh

c) 180 V d) 1220 V

c) loop

Q17. Which instrument is used as the null difference in the wheatstone bridge?

a) galvanometer b) voltmeter
c) ammeter d) multi meter

Q18. If the flow of electric current is parallel to the magnetic field, the force will be ______

a) Half of the original value
b) Maximum
c) Infinite
d) Zero

Q19. Which of the following cannot be computed Using Biot Savart law?

a) Magnetic field intensity
b) Magnetic field density
c) Permeable
d) Electric field intensity

Q25. If the frequency of the AC source in a series LCR-circuit is increased, how does the current in the circuit change?

a) Decreases then increase
b) Increases then decrease
c) Becomes zero
d) Remains constant

Q26. The Electromotive force (EMF) developed by the generator depends upon

a) Area of rotating wire
b) length of rotating wire
c) Radius of wire
d) size of magnet

Q27. Find the true statement.

a) Displacement current and conduction

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a) Displacement current and conduction current are never equal
b) The current that flows through connection wires is called conduction current
c) During charging of the capacitor, in the

Using B...

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b) Magnetic field density
c) Permeable
d) Electric field intensity

Q20. Calculate the magnetic field intensity (tesla) due to a toroid of 50 turns, 2A current and a radius of 159mm.

a) 50 b) 75
c) 200 d) 100

Q21. The material with magnetic susceptibility negative and small are called as________

a) Diamagnetic b) Paramagnetic
c) Ferromagnetic d) None of the above

Q22. Identify the diamagnetic material

a) Silicon b) Germinium
c) Silver d) Cobalt

Q23. Which among the following is true about Faraday's law of Induction?

a) An emf is induced in a conductor when it

c) Radius of wire
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a) Displacement current and conduction current are never equal
b) The current that flows through connection wires is called conduction current
c) During charging of the capacitor, in the connection wires, conduction current is discontinuous and displacement current is continuous
d) During charging of the capacitor, in the gap between the capacitor plates, conduction current is continuous and displacement current is discontinuous

Q28. Is the ratio of frequencies of UV rays and IR rays in the glass more than, less than or equal to1?

a) Insufficient data b) Equal to 1
c) Less than 1 d) More than 1

Q29. X-Rays are not used in

c) Silver
d) Cobalt

Q23. Which among the following is true about Faraday's law of Induction?

a) An emf is induced in a conductor when it cuts the magnetic flux
b) An emf is induced in a conductor when it moves parallel to the magnetic field
c) An emf is induced in a conductor when it moves perpendicular to the magnetic field
d) An emf is induced in a conductor when it is just entering a magnetic field

Q24. Which of the following is found using Lenz's law?

a) Induced emf
b) Induced current
c) The direction of induced emf
d) The direction of alternating current

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Q29. X-Rays are not used in ________

a) Photographic film
b) Photocells
c) Geiger tubes
d) Ionization Chamber

Q30. Which of the following statements is true for total internal reflection?

a) Light travels from rarer medium to denser medium
b) Light travels from denser medium to rarer medium
c) Light travels in water only
d) Light travels in the air only

Q31. Zeroth law of thermodynamics helped in the creation of which scale?

a) heat energy
b) pressure
c) internal energy
d) temperature

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Q32. The first law of

SET-D

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Q29. X-Rays are not used in

a) 1-real gas, 2-ideal gas
b) 1-ideal gas, 2-real gas
c) Both are for an ideal gas at different temperatures
d) Their graphs cannot intersect

Q35. How many particles are present in one mole of any substance?

a) 6.022×10^{22}
b) 6.022×10^{23}
c) 6.022×10^{21}
d) 6.022×10^{24}

Q36. A simple pendulum of length l and with a bob whose mass is m is moving along a circular arc of angle θ in a vertical plane. A sphere of mass m is placed at the end of the circle. What momentum will be given to the sphere by the moving bob?

a) Infinity
b) Constant
c) Unity
d) Zero

...ing SHM with amplitude

c) electric flux d) electr...

Q41. Two charges q_1 and q_2 exert some amount of force on each other. What will happen to the force on q_1 if another charge q_2 is brought close to them?

a) the force will decrease
b) the force will increase
c) the force will remain same
d) the force may increase or decrease

Q42. The direction of electric field created by a negative charge is__________

a) direct outward
b) may be outward or towards
c) circular shape
d) direct towards the charge

Q43. The electrostatic potential on the perpendicular bisector due to an electric dipole is ________

a) 1
b) infinite
c) zero
d) negative

Q44. The drift velocity does not depend upon ________

a) The cross-section of the wire
b) The length of the wire
c) The number of free electrons
d) The magnitude of the electric field

mass m is placed ... momentum will be given to the sphere by the moving bob?

a) Infinity
b) Constant
c) Unity
d) Zero

Q37. A particle is undergoing SHM with amplitude 10cm. The maximum speed it achieves is 1m/s. Find the time it takes to reach from the mean position to half the amplitude.

a) π/60 s
b) π/30 s
c) π/15 s
d) π/40 s

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b) The length of the wire
c) The number of free electrons
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Q45. Which one of the following is the practical unit of power?

a) horse power
b) watt
c) kilowatt
d) kilojoule

Q46. Farad is the unit of ______

a) Luminosity
b) Wavelength
c) Permittivity
d) Inertia

Q47. The smallest value which is measured using an instrument is known as ______

a) Absolute count b) Least count
c) Round off value d) Minimum count

Q48. For the motion with uniform velocity, the slope of velocity- time graph is equal to ______

a) 1m/s b) zero
c) initial velocity d) final velocity

Q49. A coin and a bag full of rocks are thrown in a gravity less environment with the same initial speed. Which one of the following statement is true about the situation?

a) both will travel with same speed

Q56. In which direction should the force be applied to balance a force in the direction of North-East direction?

a) South b) West
c) North -East d) South-West

Q57. The value of acceleration due to gravity of earth at the equator is less than that of the poles due to ______

a) shape and rotation of the earth
b) mass of the sun
c) mass of the earth
d) mass of the moon

Q58. A player throws a ball upwards with an initial

c) initial velocity d) final velocity

Q49. A coin and a bag full of rocks are thrown in a gravity less environment with the same initial speed. Which one of the following statement is true about the situation?

a) both will travel with same speed
b) the bag will travel faster
c) the coin will travel faster
d) bag will not move

Q50. The angular velocity of a body moving with a constant speed 'V' in a circle of radius 'r' is given by:

a) V^2/r b) Vr
c) $\frac{V}{r}$ d) $\frac{r}{V}$

Q51. Inertia of motion of a body depends on ______

a) mass b) velocity
c) volume d) acceleration

Q52. The centrifugal force always acts ______

a) towards the centre
b) in tangential direction

due to _____

a) shape and rotation of the earth
b) mass of the sun
c) mass of the earth
d) mass of the moon

Q58. A player throws a ball upwards with an initial speed of 29.4m/s. What is the direction of acceleration during the upwards motion of the ball?

a) Upwards
b) Diagonal
c) Projectile motion
d) Vertically downwards

Q59. A lap joint is always in ________ shear:

a) single b) double
c) both a and b d) none of the above

Q60. The point when the fluid comes to rest is called as __________

a) stagnation point b) rest point
c) viscous point d) boundary layer point

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a) towards the centre
b) in tangential direction
c) away from the centre
d) outside of the plane of motion

Q53. According to the work energy theorem, total charge in the energy is equal to the ______
a) half of the total work done
b) total work done added with frictional losses
c) square of the total work done
d) total work done

Q54. A stone tied with a string is rotated in a vertical circle. The minimum speed with which the string has to be rotated______
a) is independent of the length of the stone
b) decreases with increasing mass of the stone
c) decreases with increasing length of the string
d) is independent of the mass of the stone

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CHEMISTRY

Q61. Dehydration of alcohol is an example of
a) addition reaction b) elimination reaction
c) substitution reaction d) redox reaction

Q62. Which of the following alcohols is the most reactive towards esterification reaction?
a) CH_3OH b) CH_3CH_2OH
c) $(CH_3)_2CHOH$ d) $(CH_3)_3COH$

Q63. When phenol is treated with excess of bromine water, it gives which of the following product?
a) m-bromophenol b) o-and p-bromophenol
c) 2,4-dibromophenol d) 2,4,6-tribromophenol

Q64. The Williamson ether synthesis produces ethers by reacting

Q54. A stone tied with a string is rotated in a vertical circle. The minimum speed with which the string has to be rotated________

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b) decreases with increasing mass of the stone
c) decreases with increasing length of the string
d) is independent of the mass of the stone

Q55. The tendency of rotation of the body along any axis is also called:

a) torque b) moment of inertia
c) moment of couple d) force

c) $(CH_3)_2CHOH$ d) $(CH_3)_3COH$

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c) 2,4-dibromophenol d) 2,4,6-tribromophenol

Q64. The Williamson ether synthesis produces ethers by reacting

a) alcohol with a metal
b) alkoxide with a metal
c) alkoxide with an alkyl halide
d) alkyl halide with an aldehyde

Q65. Identify the correct IUPAC name of CH_3-CH=CH-CHO.

a) But-2-enal b) 2-Butenal
c) Buten-2-al d) Butenal

...type of reaction takes place upon

Q74. Nucleic acids are a polymer of nucleotide monomeric units. Each nucleotide consists of

a) base-sugar-OH b) sugar-phosphate
c) base-sugar-phosphate d) base-phosphate

Q75. Wh... deficiency

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a) But-2-enal b) 2-Butenal
c) Buten-2-al d) Butenal

Q66. What type of reaction takes place upon treatment of a ketone with HCN to form a cyanohydrin?

a) Nucleophilic addition
b) Nucleophilic substitution
c) Electrophilic addition
d) Electrophilic substitution

Q67. What is the correct IUPAC name of the compound $CH_3CH=CHCH=CHCOOH$?

a) Hexenedioc acid
b) Hexa-2,4-dienoic acid
c) Penta-1,3-dienioc acid
d) Pentenedioc acid

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c) base-sugar-phosphate d) base-phosphate

Q75. Which of the following vitamin deficiency causes Beriberi?

a) Vitamin B1 b) Vitamin B2
c) Vitamin B6 d) Vitamin B12

Q76. Select the correct IUPAC name of neopentane.

a) 3, 3-dimethylpropane
b) 1, 2-dimethylpropane
c) 2, 3-dimethylpropane
d) 2, 2-dimethylpropane

Q77. Hyperconjugation involves the delocalisation of __________.

a) σ bond orbital

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Q68. The acid with the lowest pKa value is:

a) CH_3COOH
b) CH_3CH_2COOH
c) $HCOOH$
d) $(CH_3)_2COOH$

Q69. What is the correct order of reactivity of the following alkyl halides towards ammonolysis reaction?

a) $CH_3I > CH_3Br > CH_3Cl$
b) $CH_3I > CH_3Cl > CH_3Br$
c) $CH_3Cl > CH_3Br > CH_3I$

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a) σ bond orbital
b) π bond orbital
c) Both σ and π bond orbital
d) None of the mentioned

Q78. Which of the following is not true about nucleophile?

a) donates an electron pair to an electrophile to form a chemical bond
b) all molecules or ions with a free pair of electrons or at least one pi bond can act as nucleophiles
c) nucleophile are Lewis acids by definition

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b) $CH_3I > CH_3Cl > CH_3Br$

c) $CH_3Cl > CH_3Br > CH_3I$

d) $CH_3Br > CH_3Cl > CH_3I$

Q70. The IUPAC name of CH_2CN-$CHCN$ CH_2CN is

a) 1,2,3-tricyanopropane

b) 3-cyanopentene-1,5-nitrile

c) Propane-1,2,3-tricarbonitrile

d) 1,2,3-Propane trinitrile

Q71. Which of the following can be produced by Gatterman reaction of diazonium salts?

a) Bromobenzene b) Fluorobenzene

c) Nitrobenzene d) Cyanobenzene

Q72. Class of ... which cannot be

c) Both σ and π bond orbital

d) None of the mentioned

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b) all molecules or ions with a free pair of electrons or at least one pi bond can act as nucleophiles

c) nucleophile are Lewis acids by definition

d) a nucleophile becomes attracted to a full or partial positive charge

Q79. What is the correct order of magnetic strength among the following elements?

a) Fe>Co>Ni>Cu b) Fe>Ni>Co>Cu

c) Cu>Ni>Co>Fe d) Cu>Fe>Ni>Co

Q80. When potassium dichromate crystals are heated with conc. HCl

a) O_2 is evolved

b) Chromyl chloride vapours are evolved

c) Cl_2 is evolved

... takes place

a) 1,2,3-tricyanopropane
b) 3-cyanopentene-1,5-nitrile
c) Propane-1,2,3-tricarbonitrile
d) 1,2,3-Propane trinitrile

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Q72. Class of carbohydrate which cannot be hydrolyzed further, is known as?
a) Disaccharides b) Polysaccharides
c) Proteoglycan d) Monosaccharide

Q73. [illegible] a ________.
a) Covalent bond b) Ionic bond
c) Metallic bond d) Hydrogen bond

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a) O_2 is evolved
b) Chromyl chloride vapours are evolved
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Q81. Which of the following is not a consequence of lanthanide contraction?
a) From La^{+3} to Lu^{+3}, the ionic radii changes from 106 pm to 85 pm
b) As the size of the lanthanide ions decreases the basic strength increases
c) The basic character of oxides and hydroxides decreases with increase in atomic number
d) The atomic radii of 4d and 5d series is similar